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Design for green lean six sigma to improve sustainability in the pharmaceutical industry – a case study

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ABSTRACT

This paper's purpose is to outline the use of the Design for Green Lean Six Sigma (DFGLSS) to evaluate bulk pharmaceutical product packaging and ascertain a path to its reduction and even elimination to improve an organisation's sustainability. The methodology employed was DFGLSS using the Define, Measure, Analyse, Design and Verify steps. The packaging type responsible for the highest sustainability impact was identified and replaced, resulting in a decrease in overall carbon emissions associated with packaging. Adopting a reuse strategy for this packaging type resulted in cost savings, further reductions in carbon emissions, and in the total weight of bulk packaging waste. This study is one of the first, to the author's knowledge, to apply DFGLSS within a pharmaceutical organisation or elsewhere and within the wider academic literature to improve sustainability. Thus, it contributes to a research gap by demonstrating how DFGLSS can be successfully applied to promote sustainability in pharmaceutical packaging and product packaging in general.

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Design for lean six sigma; pharmaceutical; DMADV; green lean six sigma; packaging; sustainability

1. Introduction

Over the past number of years, there has been a rapidly increasing trend of improving sustainability in the pharmaceutical industry (Milanesi et al., 2020). Furthermore, in both the United States and the European Union, there is a legal obligation for companies to report sustainability targets and progress (Bade et al., 2024). Many pharmaceutical companies have now aligned their sustainability goals with those of the Paris Agreement of the United Nations Framework Convention on Climate Change (United Nations Framework Convention on Climate Change Secretariat, 1992). This has also placed greater pressure on pharmaceutical companies to comply with the EU Directive

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on single-use plastics (Rudd-Clarke & Lundy, 2023). Consequently, now more than at any other point in time, pharmaceutical companies are focused on sustainability goals.

The pharmaceutical company where this research takes place has a target of zero waste to landfill by 2035. Focus on these targets has highlighted that the product currently manufactured by the pharmaceutical company is packaged and shipped in single-use packaging. A reduction in single-use packaging must occur for the company to meet both regulatory and corporate sustainability targets. However, project implementation and improvement are particularly challenging in a heavily regulated environment such as the pharmaceutical industry (Trubetskaya, McDermott, & McGovern, 2023) (McDermott et al., 2022). This is because processes and supplied parts are subject to validation and approvals, and anything that alters an existing process validation or product approval may affect marketing authorisation. The organisations may then be subject to regulatory submissions and revalidation, which take time (Chatterjee, 2014; McGrane et al., 2022).

Lean six sigma has been utilised within many sectors and within the pharmaceutical industry to aid problem-solving and improve productivity and quality (Byrne et al., 2021). The lean six sigma methodology has been successfully used in the pharmaceutical industry to reduce operational costs, increase the production capacity, improve the results of performance indicators, increase manufacturing flexibility, reduce lead time, minimise waste, and improve production process robustness (Silva et al., 2023). Increasingly, the design for lean six sigma (DFLSS) has been utilised to aid with new processes or product design when simply improving a process will not achieve the desired objectives (Cronemyr, 2007; Trubetskaya, McDermott, & Ryan, 2023).

The combination of Green with Lean Six Sigma (LSS) to become Green Lean Six Sigma (GLSS) has aided having an environmental focus of producing a high-quality, low-cost, and environmentally friendly product/process and has demonstrated its effectiveness in improving sustainability, reducing waste, lowering greenhouse gas emissions, and minimising water usage (Ahmad & Khan, 2022; Duarte & McDermott, 2024). When aiming to meet sustainable targets, GLSS should always be utilised in combination (Garza-Reyes et al., 2014). GLSS and, in particular, environmental value stream mapping are key components in the achievement of sustainable manufacturing (Gholami et al., 2021).

There is limited research on GLSS in Pharma for sustainability and very sparse applications of Design for Green Lean Six Sigma (DFGLSS), if any, on sustainability in the pharmaceutical industry. However, having to comply with regulations such as the EU Directive on Single-Use Plastic Packaging has increased packaging as a priority. Kanimozhi et al. (2019) explores the negative impacts of pharmaceutical packaging waste, and potential solutions are examined. Furthermore, Salmenperä et al. (2022) discuss lessons that can be learned from the food industry, such as the successful implementation of circular packaging.

Sustainability has 3 pillars, environmental, social, and economic (Duić et al., 2015), and while all of these can be improved by projects, this study will focus more on the environmental sustainability pillar, which involves managing resources and reducing pollution. However, there is a lack of research relating to the use of GLSS to achieve sustainability in pharmaceutical packaging and DFGLSS applications in this area. Thus, to fill this gap, this research aims to use the DFGLSS methodology to reduce/eliminate

single-use bulk pharmaceutical product packaging. Thus, the research question (RQ) and purpose of this research are as follows:

> Can Green Lean Six Sigma and DFLSS methodology be combined to design a new bulk pharmaceutical product packaging process/product and eliminate waste from a sustainability perspective?

The literature review is presented in section 2, and the methodology is detailed in section 3. Section 4 outlines the results from this research, implementing DFGLSS to reduce/eliminate single-use bulk pharmaceutical product packaging. Lastly, section 5 discusses the results and section 6 puts forth conclusions.

2. Literature review

2.1. Lean six sigma in the pharmaceutical industry

According to several studies, as compared to other types of industry, the pharmaceutical industry has been slow in adopting the principles of operational excellence (Bellm, 2015; O'Mahony et al., 2023). Despite this, a study by McDermott et al. (2022) found that of the Pharma multinationals surveyed, over 95% had an Operational Excellence program. However, the same study found that 45% of participants surveyed believed that regulatory compliance was a barrier to continuous improvement in the pharma sector. Any change within a pharmaceutical manufacturing facility is challenging due to regulatory restrictions (Byrne et al., 2021; Duggan et al., 2022; O'Mahony et al., 2023).

However, several authors have demonstrated the successful application of LSS within the pharmaceutical industry. For example, Byrne et al. (2021) utilised LSS to identify the cause of tablets breaking in a pharma production line and avoid over 20,888 minutes of manufacturing packaging-line downtime.

2.2. Design for lean six sigma

Design for Lean Six Sigma is particularly suited for a new process design, especially when the current process requires more than improvement (Sodhi, 2020).

The structured methodology of Define, Measure, Analyse, Design and Verify (DMADV) allows for the development of a new design of a product, process or service and verification of that design (Chandan et al., 2022a). It differs from the Lean Six Sigma's Define, Measure, Analyze, Improve and Control (DMAIC) methodology in that DMADV is utilised to design a process 'right first time' by starting anew rather than by aiming to improve it repeatedly via DMAIC (Cronemyr, 2007). The 'design' step is integral to the difference where a new process or product is put forward rather than by simply focusing on the 'I' step of improving the existing product or process (Chandan et al., 2022b)

There are sparse applications of DMADV in the literature related to the Pharma industry. However, Shannon et al. (2023) utilised DFLSS to design a Total Productive Maintenance (TPM) program in an Irish pharmaceutical manufacturing plant. Utilising DMADV, a maintenance framework was designed to optimally incorporate TPM, reliability and operational excellence with an emphasis on Overall Equipment Efficiency

(OEE), realizing a 33% reduction in planned maintenance activities and a 70% reduction in corrective maintenance; cleaning for maintenance was reduced by 50%; the pilot area where implemented had its OEE increased by 20%; and the plant availability increased by over 200 hours. Silva et al. (2023)'s research was based on interviews with pharmaceutical company managers in Brazil and they found that the lean six sigma methodology can reduce operational costs, increase production capacity, improve the results of performance indicators, increase manufacturing flexibility, reduce lead time, minimise waste, and improve production process robustness. Equally, leadership involvement and commitment are key to the implementation and sustainability of a Lean Six Sigma Program. This study highlighted factors that hinder better results of Lean Six Sigma use in the pharmaceutical industry. The most important of these is adherence to pharmaceutical regulatory guidelines.

The DFLSS structured DMADV methodology is explored in **2 separate papers** 'Application of Design for Lean Six Sigma to strategic space management' and '**the use of Design for Lean Six Sigma in the medical device industry**' related to the highly regulated MedTech industry (Ó Longaigh et al., 2023) (Trubetskaya, McDermott, & McGovern, 2023). The selection of DMADV is discussed in both papers as it is more suited to a complete process redesign or a new process. Implementation of DMADV methodology provided valuable results to both Medtech organisations under study. In the Trubetskaya, McDermott, and McGovern (2023)'s study, there was a resulting 15% new space for that area, with opportunities identified to free up 44.7% of the total manufacturing floor space and realise over €2.2 million in cost savings from applying DMADV. In the O'Longaigh et al. (2023)'s study, there was an optimisation of manufacturing space and capacity in a facility that was running out of space, and LSS improvement simply could not fix the problem, calling for a complete process redesign. **The project resulted in increased space usage of approx. 38000 sq. ft. equating to a hard cost saving of over $13 million for the organisation and a cost avoidance of $22 million.**

2.3. Green and lean six sigma for sustainability

Antony et al. (2023) described the integration of 'Green' or an environmental improvement focus with LSS as the '2nd' evolution of LSS. Ahmad and Khan (2022) discussed the merits of combining Green, Lean and Six Sigma methodologies to achieve sustainability. A combined approach of GLSS overcomes any limitations of either approach utilised on its own and results in a high-quality, low-cost, environmentally friendly product/process. According to Lim et al. (2014) and McDermott et al. (2024). Organisations that embrace GLSS now may reap greater benefits in aiding regulatory compliance (Lim et al., 2014) than those that are forced to scramble later to implement initiatives in response to ever-tightening regulatory requirements.

Lean tools such as value stream mapping can be successfully integrated with green measures and key performance indicators. Gholami et al. (2021) demonstrated the use of environmental value stream mapping in conjunction with DMAIC to achieve savings of 28% in chemical consumption and 21% in energy consumption in a non-pharma manufacturer. Thorne (2021) utilised Green LSS to assess carbon footprints in an AstraZeneca pharma site in Sweden. He utilised 'E'

or environmental value stream mapping and Kaizen to identify areas that would collectively decrease the carbon footprint of the value stream by an estimated 65% when implemented. In order to implement a LSS-Green management system, it is critical to include and integrate characteristics and principles that aid the business results and provide value to the end customer (Duarte & McDermott, 2024). The food industry has led the way and proven the effectiveness of GLSS by applying this methodology to the dairy industry and food processing industry to reduce food waste, reduce greenhouse gas emissions and reduce the environmental impact of the sector (Costa et al., 2018; Trubetskaya, McDermott, & McGovern, 2023)

2.4. Sustainability in pharmaceutical packaging

The pharmaceutical industry plays an essential positive role in worldwide healthcare; however, its methods of production contribute to environmental damage (Baltruksa et al., 2023). The ongoing global impacts of the climate crisis have resulted in changes in EU legislative and pharmaceutical corporate policy. The EU Directive on single-use plastics details that, where possible, single-use plastics should not be used. Where this is not possible, they should be, at a minimum, reusable and always recyclable. Furthermore, many pharmaceutical industries have a target of zero waste to landfill by 2035 (Drug, Chemical & Associated Technologies Association, 2024). However, 'embedding environmental sustainability into the pharma industry's DNA is a momentous challenge' (Taylor et al., 2022).

However, pharmaceutical companies are in the early days of fulfilling their sustainability responsibilities. Research conducted by García González (2020) examined pharmaceutical companies' perception of the EU Directive on Single-Use Plastic Packaging and the EU Scheme on Producer Responsibility. The research showed that pharmaceutical companies are not specifically aware of these regulations despite the fact that the industry is not excluded from the scope of the regulations. It concludes that further research is needed to solve the issue of single-use plastics in the pharmaceutical industry. Furthermore, the focus should be on high areas of criticality/impact and use risk assessment.

Although an uncomfortable reality, pharmaceutical packaging results in a large amount of waste in landfills. The research conducted by Kanimozhi et al. (2019) discusses the negative environmental impact of pharmaceutical packaging from an Indian context. The research also details the various regulatory requirements associated with pharmaceutical packaging. Different types of packaging waste are discussed, as well as environmentally friendly options such as reuse, recycling, and bioplastics. The research concludes that greater investment in research and packaging design is required to meet global requirements such as sustainability.

To meet sustainability goals, concepts such as the circularity of pharmaceutical packaging must be examined. A study conducted by Salmenperä et al. (2022) discussed the pharmaceutical product packaging value chain. Barriers to achieving circularity of pharmaceutical packaging are analysed in particular legislation, rigid practices, lack of stakeholder interaction and lack of information. This rigid legislation around pharma particularly impacts manufacturing, packaging, procurement, and logistics. Of note are

the lessons that can be learned from the food industry's successful implementation of circular packaging (De Bernardi et al., 2023).

2.5. Research gap/conclusion

A selection of relevant related literature referenced above from the Pharmaceutical and other sectors is summarised in Table 1 below, which has shown that LSS, GLSS and DFLSS have been successfully applied in several industries and, in some cases, but on a more limited basis within the pharmaceutical industry. The literature review found that stakeholder engagement is crucial to the success of a project (Ó Longaigh et al., 2023; Shannon et al., 2023; Trubetskaya, McDermott, & Ryan, 2023). Additionally, LSS has been used to achieve sustainability goals but only through the integration of Green (Baltruksa et al., 2023; Gholami et al., 2021; Taylor et al., 2022). Furthermore, sustainability in pharmaceutical packaging was reviewed, and the challenge of regulatory restrictions was noted (García González, 2020; Salmenperä et al., 2022). Similarly, the challenges of regulatory compliance or the risk of noncompliance with regulations has been referenced in the literature related to both GLSS and LSS deployment in the pharma industry (Byrne et al., 2021; McDermott et al., 2022). However, it has been found that there is a research gap in relation to the use of both DFLSS and DFGLSS to aid the achievement of sustainability targets in pharmaceutical packaging. The gap is simply not that there is little research on the area to date but rather a lack of evidence of the applicability of DFGLSS in meeting sustainability and environmental targets and goals. Without such research studies, the pharmaceutical industry cannot benchmark and leverage what has been learned from success stories to advance sustainability practices.

3. Methodology

3.1. Background to the research study

This research project was conducted at a pharmaceutical manufacturing facility in Europe. The company manufactures bulk active pharmaceutical ingredients (API) and bulk drug products (tablets and capsules). These bulk products are packed in bags and placed inside drums/totes for transport. 100% of the drums/totes used for shipping bulk products go to waste. As per the research gap identified in the literature review, there is a lack of research relating to the use of DFGLSS to achieve sustainability in pharmaceutical packaging. Therefore, this research project aims to reduce/eliminate single-use bulk pharmaceutical product packaging using GLSS and DLSS methodology. A case study approach was deployed to help examine these aims. This approach was deemed the best method to explore the adoption of DFGLSS deployment at an individual level in the specific context of a single Pharma organisation. Case studies are particularly appropriate when the issue under investigation is exploratory and when there is an absence of knowledge on the subject, as in this area (Yin, 2016). In this case, the single-site case study is appropriate as the area under investigation is unique, critical and revelatory (Dubé & Paré, 2003). The pharmaceutical industry can be secretive by nature and reluctant to release any information that may put them at a disadvantage to their competitors (Cavallaro et al., 2020). Drug patent and trade secret laws strongly

Table 1. Literature review summary (source: authors own work).

Author	Scope	Title	Item Type	Publication Year	Key Findings
Baltruksa et al. (2023)	Environmental damage caused by the pharmaceutical industry.	Strengthening sustainability in the pharmaceutical sector.	Journal Article	2023	Methods of production used by the pharmaceutical industry cause environmental damage.
Taylor et al. (2022)	Sustainability in the pharmaceutical industry.	Embedding environmental sustainability into pharma's DNA	Report	2022	Strong leadership is needed for the difficult challenge of embedding sustainability into the pharmaceutical industry.
Byrne et al. (2021) Ó Longaigh et al. (2023). Shannon et al. (2023)	The use of Lean Six Sigma in the pharmaceutical industry and the use of DMAIC. The use of Design for Lean Six Sigma in the medical device industry and DMADV. A total productive maintenance & reliability framework for an active pharmaceutical ingredient plant utilising design for Lean Six Sigma	Applying Lean Six Sigma Methodology to a Pharmaceutical Manufacturing Facility: A Case Study Strategic facility & space planning utilising Design for Lean Six Sigma. Applying Design for Lean Six Sigma Methodology to a Pharmaceutical Manufacturing Facility: A Case Study	Journal Article Journal Article Journal Article	2021 2023 2023	A demonstration of the successful implementation of Lean Six Sigma in a medical device manufacturing facility to achieve savings in downtime. The project demonstrates the application of Design for Lean Six Sigma and a structured Define, Measure, Analyse, Design, and Verify methodology in designing and implementing a process that enables the case study manufacturing site to improve its space utilisation and free up space. A case study involving Design for Lean Six Sigma phases of Define, Measure, Analyse, Design, and Verify was utilised to build an API site TPM pilot program
Molina Cordero (2022)	The use of Lean Six Sigma in the pharmaceutical industry and the use of DMADV.	Designing a Standardized Process Flow for Regulatory R&D Service Center: A Design for Lean Six Sigma DMADV Case Study	Journal Article	2022	DMADV should be used for a process redesign or a new process. Accurate data, particularly historical data, for the analysis phase is crucial. This helps ensure the accurate identification of key process variables.
Silva et al. (2023)	The use of Lean Six Sigma in the pharmaceutical industry	Analysis of Lean Six Sigma Use in Pharmaceutical Production	Journal Article	2023	Lean Six Sigma methodology can reduce operational costs, increase production capacity, improve performance indicator results, increase manufacturing flexibility, reduce lead time, minimize waste, and improve production process robustness. Strong leadership is essential, and pharmaceutical regulations can be a barrier.
Trubetskaya, McDermott, and Ryan (2023)	The use of Lean Six Sigma in the medical device industry and the use of DMADV.	Application of Design for Lean Six Sigma to strategic space management	Journal Article	2023	DMADV is most suited to a process redesign or new process. Stakeholder engagement is crucial at all stages; otherwise, a new process can be perceived by end users as another hurdle to jump.

(*Continued*)

Table 1. (Continued).

Author	Scope	Title	Item Type	Publication Year	Key Findings
Ahmad and Khan (2022)	The use of Lean Six Sigma to achieve sustainability.	The combined implementation of Green, Lean and Six Sigma approaches for achieving environmental sustainability –A Review.	Journal Article	2022	A combined approach of Green Lean Six Sigma overcomes any limitations found in just using Lean or Six Sigma or Green. This results in a high-quality, low-cost, environmentally product/process.
Nagadi (2022) Gholami et al. (2021)	The use of Lean Six Sigma to achieve sustainability. The use of Lean Six Sigma to achieve sustainability.	Implementation of green, lean and Six Sigma operations for sustainable manufacturing. A Review The application of Green Lean Six Sigma	Journal Article Journal Article	2022 2021	The research details a road map to implement a Green Lean Six Sigma framework successfully. Barriers to successful implementation of Green Lean Six Sigma include lack of stakeholder engagement, lack of environmental knowledge, failure to fully use and integrate all 3 methodologies. Green measurables should be included in value stream maps when applying a Green Lean Six Sigma approach. Stakeholder involvement is key to avoiding uncertainty during key decision-making.
Garza-Reyes et al. (2014)	The use of Lean Six Sigma to achieve sustainability.	The application of Green Lean Six Sigma	Journal Article	2014	There are limitations to using Green Lean methodologies to achieve sustainability goals and operational improvements. Using an integrated approach of Green Lean Six Sigma avoids any of the limitations associated with a singular approach.
García González (2020)	Sustainability in Pharmaceutical Packaging.	Exploring The Effects of Plastic-Free Packaging In The Pharmaceutical Industry	Journal Article	2020	Pharmaceutical companies are not specifically aware of their legal responsibilities regarding the EU Directive on Single Use Plastic Packaging and the EU Scheme on Producer Responsibility. Further research is needed on single-use plastics in the pharmaceutical industry, with a focus on high-risk areas.
Kanimozhi et al. (2019) Salmenperä et al. (2022)	Sustainability in Pharmaceutical Packaging. Sustainability in Pharmaceutical Packaging	Issues in Modern Pharmaceutical Packaging - An Indian Perspective Increasing the Circularity of Packaging along Pharmaceuticals Value Chain", Sustainability	Journal Article Journal Article	2019 2022	Single-use pharmaceutical packaging has a negative impact on the environment. Greater investment is needed in alternatives such as reuse, recycling, and bioplastics. Implementing circularity in the pharmaceutical industry could be a potential solution to single-use packaging. However, the industry's rigid regulations act as a barrier. The food industry's adoption of circular packaging can provide lessons.

discourage the willingness of the industry to publicly disclose details about their technologies, approaches, and personal stories (Murphy, 2018; Umemura, 2013). Thus, any opportunity to carry out research within the pharmaceutical sector is an opportunity to add to the state-of-the-art literature.

3.2. Design for green lean six sigma

The inclusion of Green in the DFLSS methodology ensures that sustainability metrics are at the core of the project (Ahmad & Khan, 2022). The DMADV framework, summarised in Table 2, is considered more appropriate as this project requires the creation of a new process (Trubetskaya, McDermott, & McGovern, 2023). Additionally, Table 2 provides an overview of each stage of the DMADV framework completion status. Data was obtained and collected via brainstorming sessions, utilising company scorecards and key performance indicator reports as well as supplier and industry-related data. The data collection was part of an exercise to gather the voice of the customer and the voice of the business data. The data was verified as being appropriate through stakeholder consensus and DMADV tools that aided in highlighting the criteria to sustainability measures. A wide array of stakeholders' input was sought, varying from suppliers right through to customers (a detailed list of stakeholders is presented in the SIPOC diagram in Section 4). The Design and Verification of a proposed new design is a key final component of the DMADV process, ensuring the new design meets customer and business needs (Cronemyr, 2007). Accordingly, a validation/verification plan was developed, ensuring that all measurables pertaining to sustainability and green targets were met. This will be followed by a pilot run, executed as per the requirements of the validation/verification plan. Change management plans can then be executed for large-scale rollout. Prior to this, however, lessons learned must be reviewed and taken into consideration.

4. Results

A DMADV methodology was used to evaluate bulk pharmaceutical product packaging from a sustainability perspective. Furthermore, changes that reduced and eliminated single-use bulk pharmaceutical product packaging were identified.

4.1. Define

At the initiation of any project, it is crucial to identify customers and their requirements. The define phase fulfils this purpose. As a first step in this project, a SIPOC (Suppliers, Inputs, Processes, Outputs, and Customers) diagram was created, as shown in Figure 1, for the pharmaceutical bulk product manufacturing site. Completing a SIPOC provided a full understanding of stakeholders, their inputs, the process, outputs from each process step, and customers.

Having completed a SIPOC, a Voice of the Customer (VOC) and Voice of the Business (VOB) exercise was executed, as shown in Table 3. At the heart of this was identifying Critical to Customer/Sustainability (CtQ/CtS) requirements. This was done by examining what customers are saying in relation to key issues. Through this process, customers' needs and wants can be understood, i.e. critical customer requirements. In

Table 2. DMADV (define, measure, analyse, design, verify) framework, phase deliverables, tools selected, purpose of each tool, target completion, and status.

DMADV	Deliverables	Tools applied	Purpose of tool
Define	Clearly defined project charter detailing the scope and objectives of the project.	(1) SIPOC (2) VOC & VOB (3) Project Charter	(1) Shows inputs and outputs for a process/project. (2) Gather the voice of the customer and the voice of the business to understand drivers. (3) Details the scope, objectives, resources, deliverables, and risks.
Measure	Identify critical to sustainability (CtS) measurables and gather data. Create a current state flowchart.	(1) CtS Tree (2) Current state Flowchart. (3) Data collection plan.	(1) Identify critical to sustainability (CtS) measurables. (2) Determine the current state. (3) Ensure the accurate collection of current state data.
Analyse	Analyse the current state to identify improvement areas.	(1) Graphical data analysis (2) Ishikawa cause and effect. (3) 5 Whys (4) Pugh Matrix	(1) Analyse all current state data. (2) Analyse all current state data, problem solve. (3) Identify areas for improvement. (4) Solution selection.
Design	Design a new process for the selected solution.	(1) Future state Flowchart. (2) Gannt chart	(1) Provides an overview of what the future state process will be. (2) Ensure timeline is in place for implementation of the new process.
Verify	Verify the new design works.	(1) Test validation plan. (2) Pilot run (3) Lessons learned and next steps.	(1) Plan in place to ensure all necessary requirements are checked during testing/validation of the new process. (2) Conduct a pilot run/runs as per the validation plan. (3) Identity lessons learned and ensured they are incorporated into any potential full implementation in next steps.

this case, as shown in Table 3, sustainability and cost were identified as the Critical Customer Requirements.

Within the project charter, a problem/opportunity statement was defined, 'The Company Sustainability policy has a target of zero waste to landfill by 2035.' The EU Directive on single-use plastics details that, where possible, single-use plastics should not be used (Duarte, 2023; Duarte & Cruz Machado, 2017). Where this is not possible, they should be, at a minimum, reusable and always recyclable. Moreover, the project charter noted that any solution must be compliant with pharmaceutical manufacturing regulatory requirements. Finally, resources were set out using the information gathered from the SIPOC.

4.2. Measure

The Critical to Customer or Critical to Quality (CtQ) requirements identified during the definition phase were transformed into CtS (critical to sustainability) measures through the use of a Critical to Sustainability (CtS) tree, as shown in Figure 2.

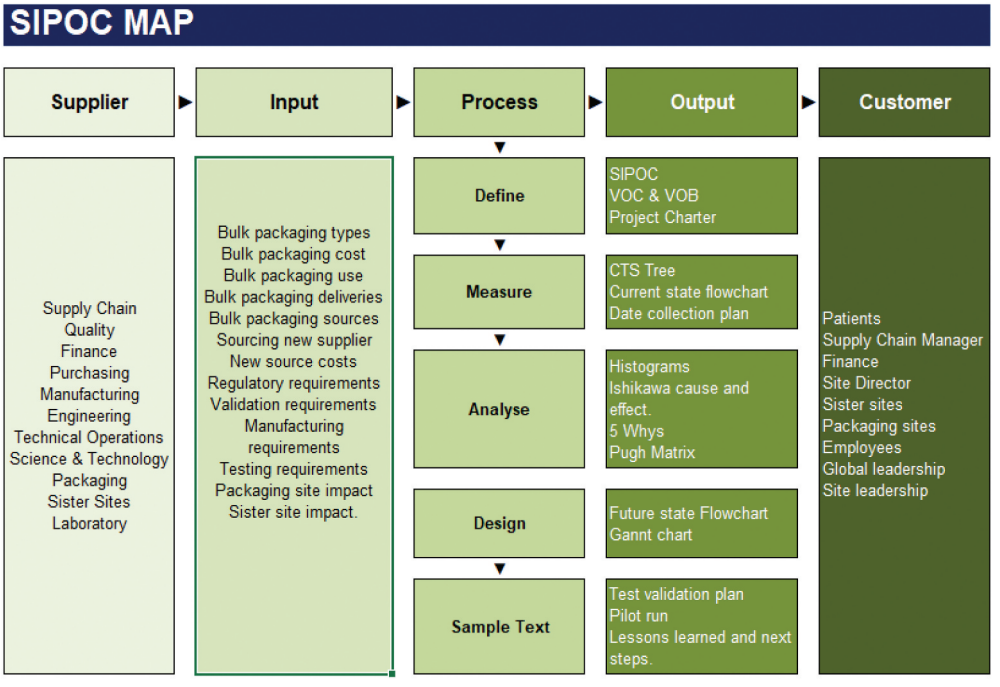


Figure 1. SIPOC (suppliers, inputs, processes, outputs, customers) used to ascertain project stakeholders, their inputs, the process, outputs from each process step and customers.

By understanding the drivers of customer and business needs, key measurables were identified. The CtS measures identified were weight to waste, CO_2 emissions due to procurement transport and cost.

With CtS priorities identified, it was then possible to create a current state flowchart, as per Figure 3. There are four types of bulk pharmaceutical product packaging used at the manufacturing site. The current state flow chart details the current process and average per delivery CTS (critical to sustainability) measurables for each of the bulk pharmaceutical product packaging types. One particular tote box measured the highest across all CtS measures. Per each delivery, this type of packaging resulted in a procurement cost of over €15000 with an associated 13.33 tonnes of carbon emissions and 2.6 tonnes of weight-to-waste ratio.

Having identified CtS measurables, a data collection plan was created and executed, as shown in Table 4. The plan laid out what data needed to be collected, from where, the timeframe, sources, formulae, search codes, stratification factors, containers, and responsibility. Additionally, the plan showed how the data to be measured related to each CtS measure.

4.3. Analyse

The data collected during the measure phase was examined in relation to all CtS (critical to sustainability) measurables, including weight to waste, CO_2 emissions due

Table 3. VOC (voice of the customer) & VOB (voice of the business) for the pharmaceutical bulk product manufacturing site in Europe. (source: authors own work).

Who is the Customer/ Business	Customer or Business	Customer Statement What are they saying?	Gathering Key Issues What is the need?	Critical Customer Requirements What do they want?
Patients	Customer	I want my medication to always be available, work and be produced ethically and sustainably.	Sustainability in pharmaceutical manufacturing	Sustainability
Supply Chain Manager	Customer	I want supply chain security.	Supply chain sustainability	Sustainability
Finance	Business	I want financial sustainability	Sustainability in costs	Cost
Site Director	Business	I want assurance of the supply of products for patients and to meet our sustainability targets.	Meeting patient needs regulatory & corporate sustainability & financial requirements.	Sustainability & Cost
Sister Sites	Customer	On-time deliveries which meet incoming inspection	Security of planned deliveries.	Sustainability
Packaging Sites	Customer	We want easy to use bulk packaging	Single-use poor ergonomic packaging	Sustainability
Employees	Customer	We want to be a part of manufacturing medication ethically and sustainably.	Personal ethics in relation to the global sustainability movement	Sustainability
Global Leadership	Business	We want consistent assurance of supply to our patients and to meet our global sustainability targets	Meeting patient needs, regulatory & corporate sustainability & financial requirements	Sustainability & Cost
Site Leadership	Business	We want consistent assurance of supply to our patients and to meet our global sustainability targets	Meeting patient needs regulatory & corporate sustainability & financial requirements.	Sustainability & Cost
Purchasing	Customer	I want supply chain security and financial sustainability	Global geopolitical issues impact suppliers	Sustainability & Cost

to procurement transport, and cost (Table 5). These CtS measures were weighed in importance by the project team utilising input from stakeholders and in particular customers.

The highest weight-to-waste ratio, carbon emissions, came from the Tote Box 35 L Lid. The 35 L Tote Box was responsible for the highest cost due to procurement and transport.

Finally, the overall impact of each CtS on each bulk packaging type was examined. 69% of the total bulk packaging weight to waste ratio came from the Tote Box 35 L Box Lid as did 58% of the total bulk packaging carbon emissions due to procurement. 45% of the total bulk packaging cost due to procurement came from the 35 L Tote Box and 43% from the Tote Box 35 L Lid.

An Ishikawa diagram (Figure 4) was used to determine the cause of the high CTS measurables for the bulk packaging type Tote Box 35 L Lid. The Ishikawa diagram exercise aided in understanding what factors were contributing to the CtS measurements via brainstorming of the most important root causes of high CtS measures.

These are as follows:

- Weight – heavier packaging results in greater weight to waste.

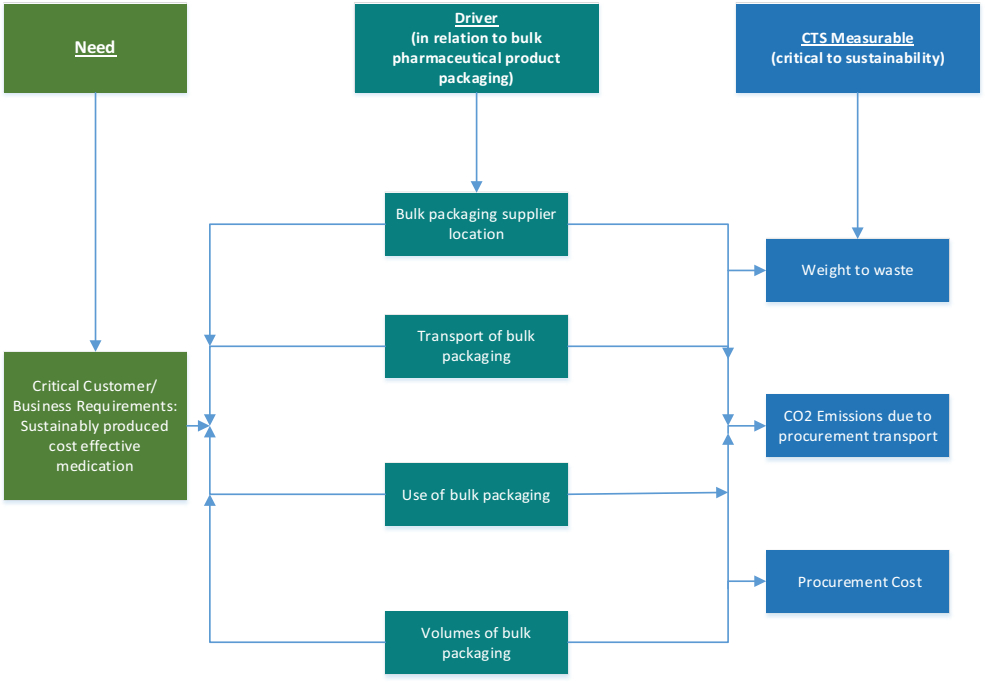


Figure 2. CtS tree, detailing customer and business needs, drivers to needs and CtS measurables.

- Material of construction – heavier materials such as plastics in this case result in greater weight to waste and cost due to procurement.
- Single use – single use of this packaging results in greater weight to waste and carbon emission due to procurement and cost due to procurement.
- Supplier location – The supplier's location causes greater carbon emissions due to procurement as they are located nearly 6000 km away.
- Sealable – The packaging must be sealable to protect the product from the environment. This is more costly, and the current design does not allow for reuse. This results in greater weight to waste and cost due to procurement.

A 5 Why analysis was conducted to delve further into each cause identified during the completion of the Ishikawa cause and effect diagram. The 5 Why analysis, as shown in Table 6, highlighted that there is no alternative packaging qualified for use, hence being restricted to the current type and source. Additionally, the bulk packaging is not designed for reuse. Consequently, it was decided that a new supplier should be sourced from a location closer to the store and provide a suitable packaging type that is reusable. Furthermore, a new process would be designed for the reuse of this bulk packaging type. This initial focus on a single packaging type would provide a maximum positive impact on CtS measurables. Moreover, this change would then provide a justification for change across multiple sites and sister sites in the network.

A replacement for the bulk packaging type Tote Box 35 L Lid, which was both reusable and from a more local supplier, was now required. A Pugh matrix (Figure 5) was used to assess five potential options, Models A – E. Ten criteria were selected and weighted in

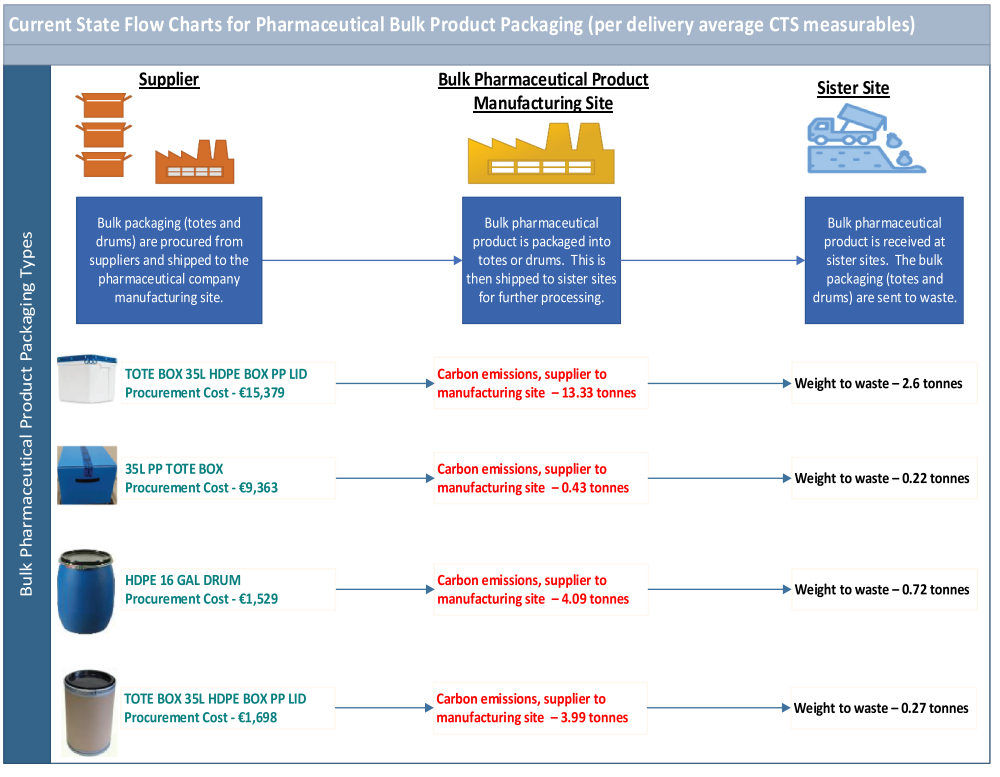


Figure 3. Current state flowchart for bulk pharmaceutical packaging including CTS measurables (source: authors own work).

accordance with importance. These included CtS measurables, meeting regulatory and manufacturing requirements, and ergonomic considerations. Model C came out as better compared to the other options available, including the current Tote Box 35 L Lid.

4.4. Design

Having selected a replacement packaging type for the Tote Box Lid, a new process could be designed for reuse of the new bulk packaging tote. To reuse the totes, a circular supply chain would need to operate between the manufacturing site and the packaging site. Once totes were sent to the packaging site from the manufacturing site and emptied, they would be stored. The manufacturing site would then have them returned for reuse as required. Accordingly, the future state flowchart was mapped out as per Figure 6. Furthermore, the future state flowchart contains anticipated savings on CtS measurables once the process is fully implemented. There was a projected 58% per year saving on the total cost of all bulk packaging. Additionally, there was a projected 79% per year saving on total carbon emissions due to the procurement of bulk packaging. Finally, there was a projected 82% per year saving on the total weight-to-waste ratio of bulk packaging.

The project was split into two phases in order to capitalise on a quick win. It was noted that simply changing from the current Tote Box 35 L Box Lid to the proposed new tote, Model C, would bring about an immediate saving in carbon total emissions due to bulk

Table 4. Data collection plan used to ascertain all CtS (critical to sustainability) measurables over a five-year period (source: authors own work).

Measurable	CTS	Source	Data Collection Mechanism	Sample	Stratification Factors	Data Container	Formula to be used	Responsible
Quantity	Weight to Waste	ERP System	ERP Report	5 years	Per delivery	Excel	N/A	Supply Chain Specialist
Supplier Location	CO2 Emissions due to procurement	Carbon footprint calculator	Route measuring	5 years	Per tote	Excel	N/A	Supply Chain Specialist
Weight per Unit	Weight to Waste	Warehouse Pharmaceutical Technician	Shipping Records	5 years	Tonnes	Excel	N/a	Supply Chain Specialist
Cost per Unit	Procurement Cost	ERP System	ERP Report	5 years	Euros	Excel	N/A	Supply Chain Specialist
CO2 Emissions per Delivery	CO2 Emissions due to procurement	Carbon footprint calculator	Carbon footprint calculator	5 years	Tonnes	Excel	N/A	Supply Chain Specialist
Weight per Delivery	Weight to Waste	ERP Warehouse	ERP report and Shipping Records	5 years	Tonnes	Excel	Weight per unit x Quantity	Supply Chain Specialist
Cost per Delivery	Procurement Cost	ERP system	ERP Report	5 years	Euros	Excel	Cost per unit x Quantity	Supply Chain Specialist

Table 5. CtS (critical to sustainability) measurables summary table by packaging type.

Bulk Packaging Type	Cost due to procurement € (euros)	Weight to waste (Tonnes)	Carbon Emissions due to procurement (Tonnes)
TOTE BOX 35L LID	474,149.40	80.16	439.89
FIBRE DRUM (82.4 LTS)	91,293.06	14.514	199.5
35L TOTE BOX	505,612.80	12.096	24.94
HDPE 16-GAL DRUM	42,140.16	8.928	94.07
Total of cumulative Bulk packaging 5 yr period	**1,113,195.42**	**115.698**	**758.4**

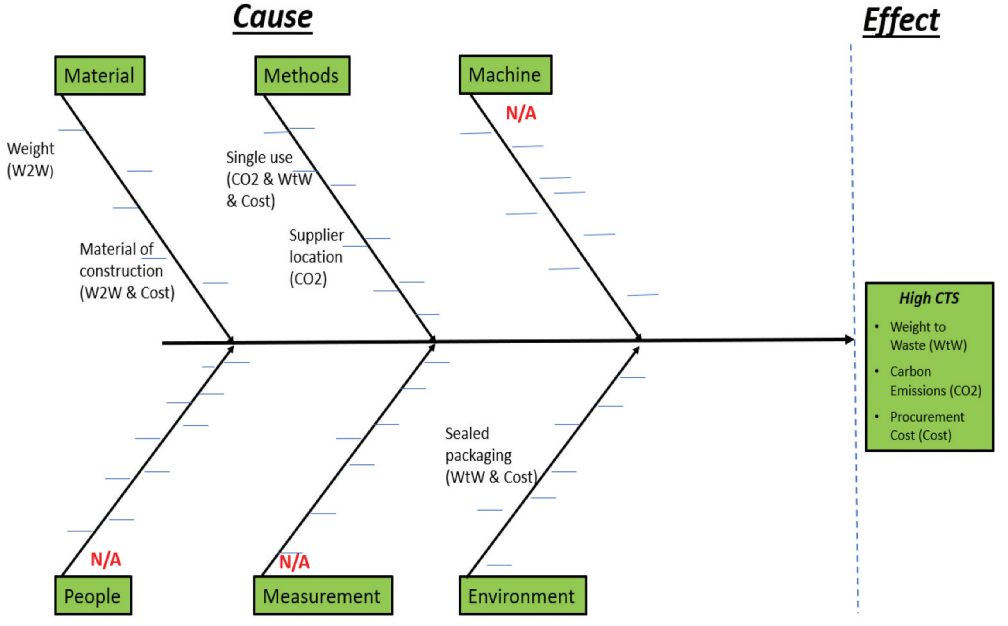


Figure 4. Ishikawa cause and effect diagram used to determine the cause of the high CTS measurables for the bulk packaging type.

packaging procurement of 66%. Additionally, the supplier qualification process and technical evaluation necessary for the new tote, would then be completed in advance of the proposed new process of reuse. The first phase involved a global cross-functional authoring of a change plan under the regulated supplier/parts quality management system to qualify the new tote and its supplier.

4.5. Verify

A robust description, reason and justification for change was documented for regulatory compliance purposes. Through this, the process validation impact was assessed. Final verification of the proposed change was completed via approval of the change plan by all impacted parties.

Table 6. Why analysis used to delve further into each cause identified during completion of the Ishikawa cause and effect diagram.

Bulk packaging weight causes a high Weight to Waste CTS measurable.	Bulk material of construction causes high Weight to Waste & Cost CTS measurables.	Single use of bulk packaging causes high Weight to Waste & Cost and CO2 CTS measurables.	Bulk packaging supplier location causes a high CO2 CTS measurable.	The requirement for sealed bulk packaging causes high Weight to Waste & Cost CTS measurables.
Why is that?	Why is that?	Why is that?	Why is that?	Why is that?
The heavier bulk packaging is the greater waste to waste is.	A heavier material of construction costs more and results in greater weight to waste.	The largest volume of bulk packaging used, is the heaviest, one of the costliest and comes from a supplier located nearly 6000 km away and is used only once.	The largest volume of bulk packaging used comes from a supplier located nearly 6000 km away.	The largest volume of bulk packaging used, is the heaviest, one of the costliest and comes from a supplier located 6000 km away and is used only once.
Why is that?	Why is that?	Why is that?	Why is that?	Why is that?
The largest amount of bulk packaging type used is also the heaviest.	The largest amount of bulk packaging type used is also the heaviest material of construction.	The products this type of packaging are used on are tablets and capsules and require a high level of protection.	The products this type of packaging are used on are tablets and capsules and require a high level of protection.	This type of packaging is used on products manufactured in largest volumes.
Why is that?	Why is that?	Why is that?	Why is that?	Why is that?
This type of packaging is used on products manufactured in largest volumes.	This type of packaging is used on products manufactured in largest volumes.	Certain regulatory requirements related to filings must be met for shipment of these products.	Certain regulatory requirements related to filings must be met for shipment of these products.	These products are tablets and capsules and require a high level of protection from the environment.
Why is that?	Why is that?	Why is that?	Why is that?	Why is that?
These products are tablets and capsules and require a high level of protection	These products are tablets and capsules and require a high level of protection	There is no alternative packaging qualified for use hence being restricted to a supplier so far away. Additionally, the largest volume bulk packaging is not designed for reuse.	There is no alternative packaging qualified for use.	Certain regulatory requirements related to filings must be met for shipment of these products.
Why is that?	Why is that?			Why is that?
Certain regulatory requirements related to filings must be met for shipment of these products.	Certain regulatory requirements related to filings must be met for shipment of these products.			There is no alternative packaging qualified for use.
Why is that?	Why is that?			
There is no alternative packaging qualified for use.	There is no alternative packaging qualified for use.			

Projected savings for CtS measurables over a five-year period and implementation costings were documented. Table 7 summarizes the projected savings reviewed and approved during the verification stage.

Consequently, lessons learned need to be evaluated prior to the execution of the second phase, the reuse of the new tote. Two key points consistently provided challenges throughout the process. Firstly, although there was an acute awareness of

Pugh Matrix for new proposed bulk packaing types.

	Criteria Definition	Model A	Model B	Model C	Model D	Model E	Importance Rating (based on team's input)
Criterion 1	Weight to waste	+	S	S	+	S	5
Criterion 2	Carbon emissions from procurement.	+	S	+	+	+	5
Criterion 3	Procurement cost.	-	-	+	+	+	5
Criterion 4	Reusability	-	-	+	-	-	5
Criterion 5	Ease of implementation	+	+	+	-	-	3
Criterion 6	Meets regulatory requirments.	+	+	+	-	-	5
Criterion 7	Meets manufacturing requirments.	-	+	+	-	-	4
Criterion 8	Meets packaging site requirments.	+	+	+	-	-	4
Criterion 9	Meets handling requirments.	+	+	+	+	+	3
Criterion 10	Ease of use sealing and opening.	+	+	+	+	+	3
Total "+"		7	6	9	5	4	
Total "S"		0	2	1	0	1	
Total "-"		3	2	0	5	5	
Weighted Total "+"	The Model is BETTER compared to current state or best in class	28	22	37	21	16	
Weighted Total "S"	The Model is ABOUT THE SAME compared to current state or best in class	0	10	5	0	5	
Weighted Total "-"	The Model is WORSE compared to current state or best in class	14	10	0	21	21	

Figure 5. Pugh Matrix used for selection of an alternate tote type.

the importance of stakeholder engagement, the time it can take to achieve that was not considered. Therefore, meetings with impacted global SMEs will be scheduled regularly for the second phase of the project. Secondly, the impact to regulatory was not fully accounted for, in that filing updates cause timeline delays. With this knowledge, more appropriate target timelines can be set for the second phase.

5. Discussion

The objective of this research was to evaluate bulk pharmaceutical product packaging from a sustainability perspective using DFGLSS to identify changes that would reduce single-use bulk pharmaceutical product packaging. Environmental sustainability aims to conserve natural resources and protect the environment in a responsible manner for the present and the future and reducing bulk packaging protects the environment. The results of this research show that the GLSS and DMADV methodology combined was successfully used to achieve the project objectives and improve 'green' or 'sutainabiliity' practices responsibly.

Firstly, the manufacturing site was able to complete a full evaluation of all bulk packaging in relation to identified CtS measurables via the Define, Measure, and Analyse steps of DMADV. Among the four types of bulk pharmaceutical product packaging examined, the Tote Box 35 L and Tote Box Lid stood out for their substantial negative impact on CtS measurables. This specific packaging option exhibited the highest

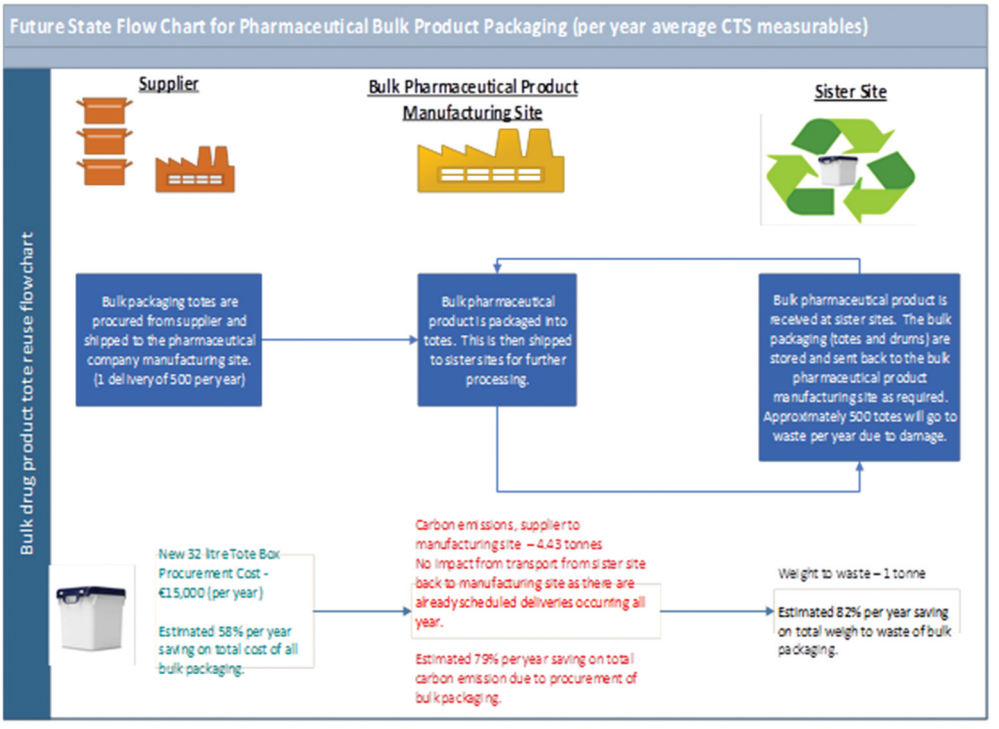


Figure 6. Future state flowchart containing CtS (critical to sustainability) measurables and projected savings.

Table 7. Summary of the projected savings reviewed and approved during the verification stage.

CtS measurable (crit ical to sustainability)	First Phase - Projected Per year saving	Second Phase – Projected Per year saving
The total cost of all bulk packaging	N/A	€138,411 = 58%
Total carbon emission due to procurement of bulk packaging.	80 tonnes = 66%	120 tonnes = 79%
Total weight to waste ratio of bulk packaging.	N/A	23.4 tonnes = 82%

carbon emissions due to its supply chain logistics, generating the most waste and having one of the highest procurement costs. Consequently, it played a major role in contributing to the overall carbon footprint and waste generation compared to the other packaging types evaluated.

Secondly, the Analysis, Design, and Verify stages of DMADV facilitated the development of changes that would reduce single-use bulk pharmaceutical product packaging and CTS measurables as a whole. It was found that by changing from the Tote Box 35 L Lid to a new tote type (Model C), sourced from a supplier located much closer, would result in a 66% reduction in total carbon emissions due to procurement. Furthermore, the

implementation of the reuse of this new tote type would result in a 58% reduction in the total cost of all bulk packaging, a 79% reduction in total emissions due to the procurement of bulk packaging and an 82% reduction in total weight to waste of bulk packaging.

The successful use of the outlined GLSS methodology is unsurprising as there is long-standing evidence since the mid 2000's of the successful combination of Green and LSS in many organisations and sectors (Duarte, 2023; Duarte & Cruz Machado, 2017; Trubetskaya, McDermott, & McGovern, 2023). Lean Six Sigma can be implemented in the pharmaceutical industry to achieve savings.

DFLSS combined with GLSS proved to be a powerful approach specifically in this case for a process redesign (McDermott et al., 2024; Trubetskaya, McDermott, & McGovern, 2023). Importantly, LSS tools such as SIPOC, VOC/VOB, CtS Tree, Flowcharts, Ishikawa diagram, 5 Whys, and the Pugh Matrix, utilised using a 'green' or 'sustainable' lens aided the evaluation of the current situation, finding a root cause, the designing of a new process and solution, and a path to pilot and verify the design and implementation (Sá et al., 2023; Thorne, 2021). The success of the DMADV approach correlates with studies in the wider pharma and other regulated industries to aid process redesign (Ó Longaigh et al., 2023; Shannon et al., 2023).

The notion of 'Green' was at the core of this research as the focus was on sustainability. One cannot measure a current state in relation to sustainability or measure improvement without selecting appropriate green measurables (Gholami et al., 2020). For this reason, in the context of this research and following Ahmad and Khan (2022), Garza-Reyes et al. (2014), and (Nagadi, 2022), green measures were the CtS measurables and were included in the current state and future state flow charts. Including green measures in flowcharts and to the fore in a project was quite alien to the organisation's project participants. This affirms the findings made by Taylor et al. (2022) that embedding sustainability into the pharmaceutical industry's DNA is a momentous challenge. The industry is largely focused on regulatory and quality compliance to the detriment of any change that may involve market authorisation risk (Speer, 2018; Umemura, 2013).

The solution for reducing single-use bulk pharmaceutical product packaging and, in turn, achieving high CtS metrics was found to be the reuse of the packaging. This concept aligns with recommendations made by Salmenperä et al. (2022) The implementation of circularity of packaging could reduce the use of single-use plastics in the pharmaceutical industry. Conversely, Salmenperä et al. (2022) also concluded that pharmaceutical regulations are a barrier to change in relation to pharmaceutical product packaging. Nonetheless, this research demonstrated that change is possible in compliance with regulations when all stakeholders are engaged in a DMADV process. The regulatory procedures and validation changes added time in project implementation in alignment with McGrane et al. (2022) study. They highlighted that an LSS project carried out by a supplier for a Medtech-regulated manufacturer (customer) took up to 6 months longer than a very similar in complexity project for an electronics manufacturer (customer). However, the results in terms of sustainability value to the business and cost savings in the case study organisation outweighed the regulatory challenges.

Kanimozhi et al. (2019) concluded that greater investment is needed in the reuse of packaging in the pharmaceutical industry and this project aided this focus. The literature review identified a lack of research relating to the use of both

LSS and GLSS to achieve sustainability in pharmaceutical packaging. Consequently, this research, conducted at a pharmaceutical manufacturing site in Ireland, now fills this gap. García González (2020) also concluded that further research is needed on single-use plastics in the pharmaceutical industry, with a focus on high-risk areas.

Moreover, this research extends the application of GLSS by utilising the DMADV methodology to design the optimal, most sustainable, and environmentally friendly solution for single-use bulk pharmaceutical product packaging. This research has unique implications for the pharmaceutical sector and other sectors by providing a benchmark study that demonstrates a defined path to measuring and achieving sustainable goals through the implementation of DFGLSS.

A limitation of the study was that it was a single-site case study restricted to non-product contact single-use bulk pharmaceutical packaging at a pharmaceutical manufacturing site in Europe. However, as there is a gap in the literature related to DFGLSS applications in the pharmaceutical sector, this study is useful for benchmarking and leveraging the learnings by others in the sector and with packaging processes.

Further research is needed to ascertain if implementing DFGLSS could reduce/eliminate other types of pharmaceutical packaging, including product-contact packaging. The project is to be expanded to carry out a sustainability evaluation pertaining to CtS measures for all raw material supply chains using the DFGLSS methodology.

6. Conclusion

The novelty of this study lies in its utilisation of GLSS and the DFLSS approach to decrease or eliminate the packaging of single-use bulk pharmaceutical products to achieve sustainability, especially the pillar of environmental sustainability. The design for green lean six sigma was introduced through the DMADV methodology that helped identify and replace the most environmentally damaging packaging. The integration of circularity principles resulted in reduction of carbon emissions and costs through the minimization of the total weight of the bulk packaging waste. The results of this study helped to establish a road map to promote sustainability in product packaging at pharmaceutical companies and many other industrial sectors. Although this study has a limitation due to it taking place only at one organisation, product packaging is common to all industrial sectors. Thus, the use of GLSS to improve sustainability can be benchmarked, and the methodology is generalisable. Moreover, this distinctive study has clear industry implications by offering the industry a clear direction for measuring and attaining sustainable objectives by adopting and merging Green Lean Six Sigma and Design for Lean Six Sigma. The study also has implications for policy as the learnings can be leveraged as proof of the importance, achievability, and impact of sustainability and environmental regulations.

Further investigation is required to determine whether the utilisation of GLSS methodology can effectively decrease or eliminate various forms of pharmaceutical packaging, including those involving product contact.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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